FoodLoop

Team Name: FoodLoop Team

Target Industry: FoodTech / Sustainability / Social Impact

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1. The Core Problem (The "Why")

Egypt's Food & Beverage sector faces a major inefficiency where large amounts of perfectly edible food are wasted while food prices continue to rise. Businesses lose revenue on unsold inventory because they lack real-time pricing and demand management tools, while price-sensitive consumers have no centralized platform to discover nearby discounted food. Traditional donation systems are also slow and inefficient, causing surplus food to spoil before reaching people in need.

2. The Solution & Value Proposition (The "What")

FoodLoop is a multi-sided AI-powered platform that connects food businesses with surplus inventory to consumers and charities in real time. Store staff scan packaged food items using a mobile camera or upload their inventory through CSV/Excel files. The system verifies product eligibility, extracts expiration information using GPT-4o Vision, intelligently adjusts prices, and publishes discounted products to nearby consumers or eligible charities.

FoodLoop's MVP focuses on packaged food products with verifiable expiration dates and does not currently support unlabeled fresh produce (such as loose fruits and vegetables), ensuring reliable expiration verification and safer automated decision-making.

Businesses can also choose how much authority they grant the AI through configurable automation levels:

- **Manual Mode:** AI provides recommendations only.
- **Assisted Mode:** AI suggests pricing while awaiting business approval.
- **Autonomous Mode:** AI automatically adjusts prices within a configurable limit.

Businesses may also completely disable automatic price changes. For food donations, FoodLoop recommends eligible items according to Egyptian food safety and donation guidelines while leaving the final approval to the business owner. Every donation decision is logged for transparency and auditing.

FoodLoop does not manage delivery operations. Customers may pick up orders directly from participating stores, while businesses may optionally arrange delivery using their own existing delivery providers.

Single most important benefit: Businesses recover revenue from products that would otherwise be wasted while consumers gain affordable access to verified food with minimal manual effort.

3. The AI Trinity Strategy (The "How")

LLM (Intelligence)

FoodLoop uses GPT-4o as its primary Large Language Model because it combines high-quality reasoning, multilingual support (Arabic and English), and strong vision capabilities within a single model.

GPT-4o is responsible for:

- Understanding expiration information extracted from product packaging images.
- Reasoning about optimal pricing decisions based on expiration time, demand, and historical sales.

- Summarizing recommendations for business owners.
- Supporting Arabic-first interactions for the Egyptian market.

RAG (Knowledge)

Rather than relying solely on the LLM's internal knowledge, FoodLoop grounds every pricing and donation decision using a Retrieval-Augmented Generation (RAG) pipeline.

The knowledge base includes:

- Egyptian food safety and donation regulations.
- Food handling and eligibility guidelines.
- Historical sales and demand patterns.
- Partner inventory information.
- Store and customer geographic data.

The RAG pipeline uses:

- **Vector Database:** Pinecone
- **Embedding Model:** OpenAI text-embedding-3-large
- **Chunking Strategy:** Documents are divided into approximately 500–800 token chunks with overlapping context to maximize retrieval accuracy.

This ensures AI decisions remain factual, consistent, and grounded in trusted business and regulatory data rather than relying solely on the LLM.

Agents (Action)

FoodLoop uses LangGraph as its agent framework to coordinate autonomous AI workflows. Unlike traditional backend services that execute static scheduled tasks, LangGraph enables AI agents to continuously observe, reason, and act while maintaining state across multiple decision steps.

Primary Goal: Maximize revenue recovery while minimizing food waste by ensuring surplus inventory is sold or donated before expiration.

Inventory Monitoring Agent

This agent continuously monitors uploaded inventory and identifies products approaching expiration. Instead of simply running scheduled database scans, it prioritizes products according to urgency and routes them into the pricing workflow.

Inventory data is synchronized through periodic CSV/Excel imports uploaded by partner businesses, allowing the AI to operate even when stores lack direct API integrations.

Price Adjustment Agent

This agent operates in a continuous Observe → Reason → Act cycle. It continuously evaluates:

- Remaining shelf life.
- Local customer demand.
- Historical purchasing trends.
- Current inventory levels.
- Previous pricing decisions.

Using GPT-4o together with the RAG knowledge base, the agent determines the optimal discount percentage needed to maximize the probability of sale.

If an item remains unsold, the agent repeats the decision cycle and updates its recommendation according to the remaining time before expiration.

The business owner always controls the level of automation, and autonomous price adjustments are limited to a configurable maximum range of ±15%, ensuring the AI operates within business-defined boundaries rather than having unrestricted control.

4. Market & Competition

Primary Users

FoodLoop serves multiple stakeholders within Egypt's food ecosystem:

- **Food Businesses/Store Owners:** Supermarkets, restaurants, bakeries, cafés, hotels, convenience stores, and grocery chains seeking to recover revenue from surplus inventory.
- **Consumers:** Price-sensitive shoppers, including students and working professionals, looking for affordable, verified food nearby.
- **Charities & NGOs:** Organizations searching for eligible food donations before products expire.

FoodLoop is designed primarily for the Egyptian market, providing an Arabic-first user experience with full English support to maximize accessibility across different user groups.

Competitive Advantage

Unlike traditional marketplaces or simple discount platforms, FoodLoop continuously monitors inventory and makes intelligent, data-driven decisions in real time. Its competitive advantages include:

- AI-powered expiration verification using GPT-4o Vision instead of relying on manual data entry.
- Dynamic pricing based on expiration time, historical demand, inventory, and location.
- Continuous autonomous decision-making through LangGraph agents rather than fixed scheduling rules.
- AI recommendations grounded in food safety regulations using a RAG pipeline.
- Business-controlled automation that allows owners to define how much authority the AI has over pricing decisions.
- Localized Arabic support tailored for Egyptian businesses and consumers.

This combination enables businesses to recover more revenue while minimizing unnecessary food waste without adding operational complexity.

Competitors

Mystery Bag

Mystery Bag primarily focuses on selling surplus bakery products and relies heavily on manual listing and fixed discounts. FoodLoop differentiates itself by:

- Verifying packaged food expiration dates using GPT-4o Vision.
- Dynamically adjusting prices throughout the product lifecycle instead of using static discounts.
- Continuously monitoring inventory to maximize recovery before expiration.

Flashfood

Flashfood specializes in large supermarkets, focusing on near-expiry items and utilizing dedicated pickup areas like the "Flashfood Zone" for quick customer collection. FoodLoop differentiates itself by

- Integrating charitable donations (Charities) directly into the same platform alongside commercial sales.
- Ensuring all donation and recovery processes automatically comply with food safety regulations using our RAG pipeline.

Phenix

Phenix offers categorized "anti-waste baskets" (such as cooking, ready-to-eat, or vegetarian baskets) to match user preferences, but managing these specific categories requires ongoing manual administrative effort. FoodLoop differentiates itself by:

- Fully automating the inventory evaluation process through our autonomous Inventory Monitoring Agent.
- Automatically identifying which items need to be dynamically priced or routed for donation without requiring continuous manual intervention, saving significant time for business owners.

Other Notable Mentions

FoodLoop's AI-driven approach also bypasses the limitations of other traditional models in the market, including **Too Good To Go** (which relies on manual entry and blind "Surprise Bags" with fixed pricing), **Karma** (which requires complex POS integrations), and **Olio** (which relies on manual data entry by volunteers and focuses primarily on free distribution without providing a dynamic revenue recovery mechanism for businesses).

Why AI Is Necessary

Food waste is a highly time-sensitive problem. Traditional software can automate predefined rules but cannot continuously reason about changing demand, expiration risk, and pricing strategies.

FoodLoop's AI agents repeatedly evaluate inventory conditions, retrieve relevant business and regulatory knowledge through RAG, and make context-aware pricing recommendations that adapt as conditions change. This enables businesses to respond to changing inventory conditions in real time while maintaining human oversight whenever desired.

Business Workflow Overview

- A business uploads its inventory through a CSV/Excel file.
- FoodLoop synchronizes the uploaded inventory with its SQL database.
- GPT-4o Vision extracts expiration information from packaged food images when needed.
- The RAG system retrieves relevant food safety regulations, historical demand, and inventory context.
- LangGraph agents determine the optimal pricing strategy or donation recommendation.

Depending on the selected automation level:

  - The business reviews the recommendation, or
  - The AI automatically applies pricing changes within the configured ±15% limit.

- Discounted products become visible to nearby consumers, while eligible donation items are suggested to registered charities.
- Customers either reserve products for store pickup, or participating businesses may arrange delivery through their own delivery services.

5. Execution Plan (High Level)

- **Core Stack** Frontend: Web dashboard for business admins using React / NextJS
- **Mobile:** Mobile app for store staff, consumers, and delivery captains using Flutter
- **Backend / AI Integration:** API layer connecting mobile, web, AI agents, and store databases using .Net (dotnet)
- **Database:** Stores food item images, product metadata, safety rules, location data, and order/delivery records using SQL
- **Observability:** Sentry for monitoring requests, latency, errors, and API costs.

- **Deployment:** Docker containers deployed on Microsoft Azure with GitHub Actions CI/CD and automated testing.
- **Security:** HTTPS, authentication, role-based access control, encryption, and audit logs protect user and business data.

- **Risk Factor:** The biggest technical challenge is achieving accurate, real-time food image recognition across diverse product labels (varying fonts, languages, lighting conditions) — errors here reduce seller trust and slow adoption. The biggest data challenge is establishing reliable, real-time inventory synchronization with partner stores that may have fragmented or non-standardized systems, which is critical for the AI pricing agents to function correctly.

6. AI Evaluation & Success Metrics

To ensure FoodLoop's AI components are reliable and measurable rather than a black box, the following metrics will be tracked throughout development and validated during a pilot phase with partner stores before full rollout:

Metric	How It's Measured	Target
OCR Accuracy (expiration extraction)	AI-extracted expiration dates compared against human-labeled ground truth on a held-out test set of product images.	≥ 95% on clear images
Retrieval Precision (RAG)	Precision@5 — of the top 5 retrieved chunks per query, the proportion judged relevant by human review.	≥ 90%
Dynamic Pricing Success Rate	Share of products entering the pricing pipeline that are sold or donated before expiration, vs. wasted.	≥ 80% recovery rate

These targets represent design goals for the pilot phase; actual performance will be measured against them once real usage data is available, and thresholds (e.g., the confidence cutoff for flagging OCR results) will be tuned accordingly.

7. Legal & Operational Clarifications

7.1 Food Safety Legal Responsibility

- **The Business** is legally responsible for the accuracy of the product data and expiration information it uploads, since the business owns and confirms this data before publication.
- **FoodLoop** is responsible for the accuracy of its AI tooling (expiration extraction and pricing/donation recommendations), not for the final decision to sell or donate a given item.
- **Mandatory human confirmation:** Even in Autonomous Mode, the business must confirm the AI-extracted expiration date before a product is published — the AI never fully bypasses business sign-off on safety-critical data.
- **Terms of Service:** This division of responsibility will be stated explicitly in FoodLoop's Terms of Service, which all partner businesses must accept before onboarding.

7.2 Pricing Agent Execution Frequency

The Price Adjustment Agent does not run on every request in true real time, which would be costly and unnecessary. Instead, it runs on a hybrid schedule:

- **Event-triggered:** triggered when new inventory is uploaded, or when an item crosses a defined expiration threshold (e.g., 48 hours or 24 hours remaining).
- **Periodic:** a review cycle every 4–6 hours for items already in the critical (near-expiration) window.

7.3 Price Reduction Limits

- **Per-cycle limit:** autonomous price changes are capped at ±15% of the current price per adjustment cycle, not a one-time unlimited drop. This is cumulative across cycles (the price can continue to decrease over time as the item approaches expiration), but each individual adjustment is bounded to protect against erratic single-step changes.
- **Price floor:** the business owner sets an absolute minimum price (e.g., based on product cost plus a minimum margin, or a fixed percentage of the original price) that the AI can never go below, regardless of how many adjustment cycles occur. If the floor is reached before an item sells, FoodLoop shifts it to a donation recommendation instead of continuing to discount.

Example: An item priced at 100 EGP with a business-defined floor of 65 EGP can be discounted in successive 15%-per-cycle steps, but the system will never let the price fall below 65 EGP — it will stop at the nearest point above the floor or route the item to donation.

7.4 AI Cost Optimization

- **Batching:** images and inventory updates are processed in batches rather than triggering a separate real-time GPT-4o call per item.
- **Caching:** OCR results are cached per product, so an item that has already been scanned is not re-sent to GPT-4o Vision on subsequent checks.
- **Tiered calls:** simple checks (e.g., "is this item approaching its expiration threshold?") are handled with plain rules/logic and do not invoke the LLM; GPT-4o is called only for genuinely complex reasoning steps, such as setting a discount percentage or evaluating donation eligibility.
- **Retrieval caching:** frequently repeated RAG queries (e.g., standard food safety rules that rarely change) are cached to reduce redundant retrieval and embedding calls.

7.5 Required Legal Documents

Store Owners (Businesses)

- **Health Certificate (شهادة صحية):** Verifies that personnel handling food are medically certified to comply with food safety regulations.
- **Commercial Registration Certificate (السجل التجاري):** Confirms that the business is legally registered and authorized to operate.
- **Tax Registration Card (البطاقة الضريبية):** Verifies that the business is officially registered with the Egyptian Tax Authority.

Charitable Organizations

- **Official Registration Decision (قرار الإشهار):** Confirms that the charity is legally established and recognized by the relevant authorities.
- **Board of Directors List (كشف أسماء مجلس الإدارة):** Identifies the organization's current governing board for verification purposes.
- **Association Bylaws (النظام الأساسي للجمعية):** Defines the charity's legal structure, objectives, and governance rules.